Reg. No. | | | | | | | | | | | [13AE61/12AE61]

SIXTH SEMESTER B.TECH/B.E UNIVERSITY EXAMINATION, JUNE – 2016

AEROSPACE DESIGN-I

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. Explain **14**

a. Critical Mach number

b. Drag divergence Mach number

c. Wing tip vortices

d. Downwash

e. Induced angle of attack

f. Effective angle of attack

g. Induced drag

2. a. Describe propeller and its principle of operation. **07**

b. A propeller is required to produce a thrust of 4000N at a flight speed of 120m/sec at sea level. If the diameter is 2.5m, estimate the nominal power that must be supplied on the basis of momentum theory. **07**

3. a. Drive the explain expression for range, condition for maximum range and maximum range for a propeller aircraft. **08**

b. Calculate the maximum range for a jet engine aircraft at sea level ISA conditions for the aircraft data given below. **06**

Max Tow=15000kg C_t=0.7 kg/kgf hour.

Fuel weight =4000kg, AR=6, $C_d=0.0012+0.06C_L^2$, S=40 sqm.

4. a. Explain any two mission profile **04**

i. air superiority mission

ii. Low level strike mission

iii. Civil transport mission

b. Calculate the initial estimate of AUW for a turbofan Civil transport aircraft for the following data; no. of passengers =100, crew=5, range =2500km, loiter=30min, C_t=0.6kg/kgf hour, $\left(\frac{L}{D}\right)_{max}=16$, Cruise speed=600Kmph. Empty weight ratio=0.5. State all the assumptions. Assume sea level ISA conditions. **10**

5.	**a.**	With neat diagram explain the landing profile for an aircraft and highlight the critical points of the profile.	**06**
	b.	Define load factor and explain pull up and pull down maneuvers. Derive expression for radius of turn and turn rate for these two maneuvers.	**08**
6.	**a.**	Explain different types of wing stall and the influence of wing taper on wing stall.	**08**
	b.	Discuss various options, its merits and disadvantages of the vertical positions of wing with respect of fuselage.	**06**
7.	**a.**	Explain how to decide on the number of engines for an aircraft.	**06**
	b.	Explain wing mounted inlet/engine and compare	**08**
		i. under the wing and	
		ii. over the wing configuration	
8.		Write short notes on:	
		a. Tail volume ratio	**3.5**
		b. Control surface sizing	**3.5**
		c. Weight and CG envelop	**3.5**
		d. Canard tail configuration	**3.5**